

funcml: Functional Machine Learning Software for R

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Abstract

funcml is an open-source R package for supervised tabular machine learning with a compact and functional workflow interface. The package provides a consistent surface for model fitting, prediction, resampling-based evaluation, hyperparameter tuning, learner comparison, model interpretation, and plug-in g-computation. Its design is intentionally explicit: formulas remain central, preprocessing is kept outside modeling calls, and return objects follow stable S3 contracts. **funcml** is intended as a lightweight alternative to larger orchestration ecosystems when users want end-to-end functionality with lower interface overhead. This paper presents the software contribution, positions the package relative to established R machine-learning frameworks, and summarizes software-quality evidence from the released source package, including documentation, testing, and contributor guidance.

Keywords: R package, machine learning software, resampling, model interpretation, causal estimation

1 Introduction

funcml is an open-source R package for supervised tabular machine learning. It was developed to make common analysis tasks easier to carry out within a single, coherent interface. In many R workflows, fitting a model, generating predictions, running resampling-based evaluation, tuning hyperparameters, comparing learners, interpreting fitted models, and estimating causal effects often involve different packages with different conventions. This is flexible, but it can also make analyses harder to read, maintain, and reproduce. **funcml** was designed to reduce that friction while keeping the modeling workflow explicit.

The package follows a formula-first style that remains close to ordinary R practice. Inputs are passed directly, preprocessing is handled outside the modeling call, and return values follow stable S3 conventions. The aim is not to replace the broader R machine-learning ecosystem, but to provide a lighter software layer for users who want a direct and readable workflow for tabular data analysis.

This article is a software paper. It describes the scope of the package, the main interface decisions, its relation to existing R frameworks, and the current state of the software implementation.

2 Software Overview and Design

The main user-facing functions in **funcml** are `fit()`, `predict()`, `evaluate()`, `tune()`, `compare_learners()`, `interpret()`, and `estimate()` [El Badisy, 2026]. Together, they cover the main stages of a super-

Table 1: Positioning of `funcml` relative to common R machine-learning frameworks.

Package	Interface style	Emphasis
mlr3 (+ extensions)	Object-oriented tasks/learners/graphs	Modular benchmarking and pipeline composition
tidymodels	Multi-package grammar	Workflow orchestration with integrated preprocessing
funcml	Compact formula-based API	Explicit fit/evaluate/tune/compare/interpret/estimate calls in one package

vised tabular machine-learning analysis.

The package is built around a small number of practical design decisions. First, the interface keeps the main analysis components visible: the data, formula, learner, resampling scheme, and optimization target are all supplied directly by the user. Second, fitted objects follow stable S3 contracts and support familiar downstream methods such as `print()`, `summary()`, and `plot()`. Third, preprocessing is kept outside the model-fitting call. This avoids hiding data preparation steps inside a workflow object and makes the modeling step easier to inspect.

The same design carries over to post-fitting tasks. Interpretation methods work directly on fitted `funcml` objects, and plug-in g-computation uses similarly direct inputs for causal estimands such as the average treatment effect [Snowden et al., 2011]. The result is a package that tries to keep the full analysis path readable from the function call itself.

3 Relation to Existing Software

R already has several mature machine-learning frameworks, including `mlr` [Bischl et al., 2016], `mlr3` and its extensions [Lang et al., 2019, Binder et al., 2021], and `tidymodels` [Kuhn and Silge, 2022]. `funcml` is not intended to compete with these frameworks on breadth. Instead, it targets a different point in the design space: a smaller, formula-based interface for users who want to run end-to-end tabular analyses without adopting a larger task, graph, or workflow abstraction.

That choice involves a trade-off. Broader frameworks provide richer infrastructure, deeper extensibility, and more elaborate pipeline systems. `funcml`, by contrast, emphasizes a shorter path from data to fitted model, evaluation results, interpretation output, and causal estimates. For users working on applied tabular problems, that narrower scope can be an advantage when the priority is to keep the workflow direct and easy to inspect.

4 Software Quality and Usability

`funcml` is released as an open-source R package under the GPL-3.0-only license and is developed through a public GitHub repository [El Badisy, 2026]. The reviewed version is `funcml` 0.7.1. The repository provides the package source code, documentation, tests, issue tracking, contributor guidance, and versioned release notes. The package also includes a README, reference manual, and vignette that document installation, core usage patterns, and the main user-facing functions. Testing is implemented through a `testthat` suite covering core components of the package, including model fitting, resampling-based evaluation, tuning, interpretation, and estimation.

5 Minimal Example

The following compact example shows how the same interface can be used across fitting, prediction, evaluation, interpretation, and causal estimation.

```
demo_dat <- funcml::arthritis
xgb_spec <- list(nrounds = 30, max_depth = 3, eta = 0.1)

fit_obj <- fit(
  status ~ age + gender + bmi + diabetes + smoke,
  data = demo_dat,
  model = "xgboost",
  spec = xgb_spec
)

pred_prob <- predict(fit_obj, demo_dat[1:5, ], type = "prob")

eval_obj <- evaluate(
  data = demo_dat,
  formula = status ~ age + gender + bmi + diabetes + smoke,
  model = "xgboost",
  spec = xgb_spec,
  resampling = cv(v = 4, seed = 42)
)

perm_obj <- interpret(
  fit = fit_obj,
  data = demo_dat,
  method = "permute"
)

ate_obj <- estimate(
  data = demo_dat,
  formula = status ~ smoke + age + gender + bmi,
  model = "glm",
  estimand = "ATE",
  treatment = "smoke"
)
```

Rather than serving as a benchmark, this example is meant to show how the package uses the same overall interface across several common analysis tasks.

6 Conclusion

funcml provides a focused software interface for supervised tabular machine learning in R. Its contribution is primarily architectural: it brings together model fitting, prediction, evaluation, tuning, learner comparison, interpretation, and plug-in causal estimation within a single formula-based workflow. The package is intended for users who want these tasks to remain explicit and

close to ordinary R analysis practice. Future development will expand learner support and continue strengthening the surrounding software infrastructure.

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